

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Microsoft, WA -- station KEPH, America/Los_Angeles
Committed pair OSM 694493127 -> 410408074
 Microsoft / Microsoft
Facade gap 146.9 m between the two halls
Weather record 42,965 real hourly records from KEPH
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise
 0.3650 C at 155 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-04-18 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. The reactive incumbent took 11 hours -- 10 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 11 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
 the incumbent broke its own switch budget 1 time(s) to stay safe
 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	8.116	18.0	7.780	1.170
01	mechanical	yes	switch budget	8.672	18.0	8.890	1.105
02	mechanical	yes	switch budget	7.550	18.0	6.670	1.065
03	mechanical	yes	switch budget	8.704	18.0	8.890	1.126
04	mechanical	yes	switch budget	7.273	18.0	5.000	1.042
05	mechanical	yes	switch budget	5.880	18.0	4.440	0.934
06	mechanical	yes	switch budget	8.940	18.0	6.332	1.340
07	mechanical	yes	switch budget	10.179	18.0	9.440	1.862
08	mechanical	yes	switch budget	12.663	18.0	13.115	1.966
09	mechanical	yes	switch budget	15.091	18.0	15.848	1.912

10	mechanical	no	dry-bulb	18.590	18.0	19.183	1.875
11	mechanical	no	dry-bulb	20.885	18.0	21.332	1.618
12	mechanical	no	dry-bulb	23.212	18.0	24.741	1.421
13	mechanical	no	dry-bulb	25.729	18.0	27.221	1.251
14	mechanical	no	dry-bulb	27.088	18.0	28.345	1.109
15	mechanical	no	dry-bulb	28.151	18.0	28.890	1.005
16	mechanical	no	dry-bulb	28.654	18.0	28.330	0.931
17	mechanical	no	dry-bulb	28.391	18.0	27.220	1.223
18	mechanical	no	dry-bulb	26.986	18.0	25.000	1.179
19	mechanical	no	dry-bulb	25.314	18.0	22.780	1.333
20	mechanical	no	dry-bulb	23.966	18.0	20.560	1.934
21	mechanical	no	dry-bulb	22.270	18.0	19.440	1.731
22	mechanical	no	dry-bulb	20.338	18.0	17.780	1.408
23	FREE	yes	-	16.751	18.0	12.220	1.316

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.116 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.672 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.549 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.705 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.273 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.880 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.940 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.179 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.663 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.091 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.590 C against a 18.0 C limit, so it fails by 0.590 C. It would take a limit 0.590 C higher, or a bound 0.590 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.885 C against a 18.0 C limit, so it fails by 2.885 C. It would take a limit 2.885 C higher, or a bound 2.885 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.212 C against a 18.0 C limit, so it fails by 5.212 C. It would take a limit 5.212 C higher, or a bound 5.212 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.729 C against a 18.0 C limit, so it fails by 7.729 C. It would take a limit 7.729 C higher, or a bound 7.729 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.088 C against a 18.0 C limit, so it fails by 9.088 C. It would take a limit 9.088 C higher, or a bound 9.088 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.151 C against a 18.0 C limit, so it fails by 10.151 C. It would take a limit 10.151 C higher, or a bound 10.151 C tighter,

to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.654 C against a 18.0 C limit, so it fails by 10.654 C. It would take a limit 10.654 C higher, or a bound 10.654 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.391 C against a 18.0 C limit, so it fails by 10.391 C. It would take a limit 10.391 C higher, or a bound 10.391 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.986 C against a 18.0 C limit, so it fails by 8.986 C. It would take a limit 8.986 C higher, or a bound 8.986 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.314 C against a 18.0 C limit, so it fails by 7.314 C. It would take a limit 7.314 C higher, or a bound 7.314 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.966 C against a 18.0 C limit, so it fails by 5.966 C. It would take a limit 5.966 C higher, or a bound 5.966 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.270 C against a 18.0 C limit, so it fails by 4.270 C. It would take a limit 4.270 C higher, or a bound 4.270 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.338 C against a 18.0 C limit, so it fails by 2.338 C. It would take a limit 2.338 C higher, or a bound 2.338 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.751 C, which is 1.249 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.316 C, and the margin is measured, not chosen: 1.231 C of group-conditional forecast error for this hour of day, plus 0.0859 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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